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MKEC 1123
ADVANCED MICROPROCESSOR SYSTEM

MILESTONE 1
Blinky

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Milestone 1: Blinky Application Setup Instruction

1. Introduction

In this milestone, a simple Blinky application was developed using the STM32 NUCLEO-F446RE development board. The purpose of this task is to verify that the STM32 development environment, firmware generation process, hardware connection, and code uploading process are functioning correctly before proceeding to the smart adaptive traffic light system project.

2. Hardware and Software Setup

The hardware used includes the STM32 NUCLEO-F446RE development board and a mini-USB cable for communication and firmware uploading. The software tools used are STM32CubeMX and STM32CubeIDE.

STM32CubeMX was used to configure the microcontroller peripherals and generate initialization code automatically, while STM32CubeIDE was used for coding, compiling, debugging, and downloading the firmware into the STM32 board.

3. GPIO and LED Configuration

The onboard user LED (LD2) on the NUCLEO-F446RE board is connected to GPIO pin PA5. In STM32CubeMX, PA5 was configured as GPIO Output mode to allow the STM32 microcontroller to control the LED directly.

This GPIO configuration is important because the same GPIO control mechanism will later be used to control the red, yellow, and green traffic light LEDs in the smart traffic light system.

4. Code Explanation

The infinite while (1) loop within main.c was navigated to insert the application logic. At line 99, the function HAL_GPIO_TogglePin(GPIOA, GPIO_PIN_5); was added to toggle the state of the designated General Purpose Input/Output (GPIO) pin driving the onboard LED. Following this, at line 103, a 2000-millisecond execution delay was implemented by adding the HAL_Delay(2000); function call, thereby establishing a two-second interval between LED state changes.

Before entering the infinite loop, several initialization functions are executed:
HAL_Init() initializes the STM32 HAL library and system timer.
SystemClock_Config() configures the microcontroller system clock.
MX_GPIO_Init() initializes the GPIO peripherals and configures PA5 as output mode.

These functions are necessary because the microcontroller cannot control hardware peripherals correctly without proper clock and GPIO configuration.

5. Related Links

Video: https://drive.google.com/file/d/1Tg8stvzxvQk4-iMYi7r6as2ijkV3tpMH/view?usp=drive_link
Github: https://github.com/leejxfke/MKEC1123-Group1-May2026